

AD9226 Module modification for VHS-Decode - ADC Capture - ~600mVp-p@50 Ohm FS

Configuration notes:

 *Target ADC input level for 600 mVp-p
 - Gain set to ~4x gain
 * R2 / R14 = 2.2 k Ω
 * R6 & R14 270ohm - Do not change
 SNR:~70db
 THD/SINAD:~98/70
 ENOB:~11

Gain table - R2 & R14 Values:

1x 560 - vin=2.5vp-p
 1.2x 680 - vin=2vp-p
 1.7x 910 - vin=1.5vp-p
 2.5x 1.2K - vin=1vp-p
 3x 1k6 - vin=0.90vp-p
 4x 2K2 - vin=0.65vp-p
 4.1x 2k4 - vin=0.60vp-p ---- Use
 5x 3k - Not recommended SNR ~60
 Selected for SNR/ENOB and lowest noise

Explanation:

Higher value rf/rg resistors, higher johnson noise.
 Analogue Devices AD8138 does not
 recommend rf >=5k, I suggest rf >=3K

Modification:

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1. R3,R5,R8,R17 - Remove
 2. R25 - Replace with 6.8nF capacitor
 3. R17 OR R3/R25 node - Add 56ohm resistor
 4. R5 - add the 0 ohm resistor taken from R25
 5. R9 & R11 - Replace with 47ohm resistor
 6. C6lpf - on the AD9226 side and across R9/11 add 160pF
 - Provides 1-pole -3db@10Mhz LPF & ADC kickback suppression
 7. R2 & R14 - Replace with 2k2 - Recommended gain
 8. C2a/b - add 4.7pF capacitor in parallel on top of R2 & R14
 - Stability and slight LPF roll off
 9. R6 & R13 Replace with 270ohm resistors
 10. C3 - 6.8nF lifted at 45 deg for series with R16 - added next
 11. R16a/b - Two 51ohm resistors in parallel, one on top
 - Add the resistors to ground side pad (AD9226 Text side)
 and lifted C3 in previous step. Provides DC offset balance.



